

Heterogeneous Decentralized Fine-Tuning of Large Language Models

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Abstract—Federated Learning (FL) enables privacy-preserving training of Large Language Models (LLMs) but struggles with system heterogeneity, particularly when clients possess varying computational capabilities. Standard aggregation methods require uniform model architectures, forcing a “lowest common denominator” approach that throttles high-end devices. We propose Subspace Projection Aggregation (SPA), a novel framework that addresses this by treating client updates as overlapping subspaces. Using Singular Value Decomposition (SVD), SPA projects high-rank updates from capable clients into optimal lower-rank approximations for resource-constrained peers. Experiments on the Yelp Review dataset using Qwen2.5-7B show that SPA achieves 63.8% accuracy, outperforming homogeneous baselines by 2.6% while reducing communication costs by 25%. Our approach effectively distills knowledge from heterogeneous sources, proving that device diversity can be an asset rather than a liability in federated LLM fine-tuning.

Index Terms—Federated Learning, Large Language Models, LoRA, Heterogeneity, SVD, Subspace Learning

I. INTRODUCTION AND MOTIVATION

Large Language Models (LLMs) have fundamentally transformed Natural Language Processing, demonstrating emergent abilities across diverse tasks [1]–[3]. However, deploying them in sensitive domains like healthcare or finance creates significant privacy issues. Regulations like GDPR [4] and HIPAA prevent centralizing sensitive data, while recent studies have shown that LLMs can memorize training data, rendering them vulnerable to extraction attacks [5], [6].

Recent incidents underscore the critical importance of privacy-preserving approaches. In 2025, the Grok Chat Leak exposed over 370,000 private conversations through centralized server vulnerabilities [7]. This is not an isolated incident. In 2023, Samsung employees accidentally leaked confidential software code by pasting it into ChatGPT for debugging [8]. Furthermore, researchers have proven that large language models can memorize and reveal private user data, including emails and phone numbers, even when safety filters are active [9]. Such breaches highlight why data governance frameworks mandate strict controls on personal information processing.

Federated Learning (FL) offers a solution by keeping data on-device [10], [11]. To make training feasible on these resource-constrained devices, Parameter-Efficient Fine-Tuning (PEFT) methods are used, with Low-Rank Adaptation (LoRA) being the most prominent technique to reduce memory usage [12]. However, a critical hurdle remains: system hetero-

geneity. In real-world networks, not all devices are equal; a high-end server may exist alongside a resource-constrained smartphone. This leads to a LoRA rank mismatch, where different devices require different model configurations to function.

A. The Heterogeneity Problem

The disparity between client devices creates a fundamental incompatibility in standard Federated Learning. Traditional aggregation methods assume that all clients share the exact same model architecture. When this assumption is broken, several critical issues arise:

Hardware Mismatch: Real-world networks are diverse. High-tier clients with 48GB VRAM can easily support a LoRA rank of 32, allowing them to learn complex features. In contrast, edge-tier devices (like smartphones or older laptops) with only 8GB VRAM are strictly limited to rank 4 or 8. If we force the entire network to use the lowest rank to accommodate the weakest devices, we waste the potential of the powerful ones.

Aggregation Failure: Standard FL relies on averaging weight updates (FedAvg). However, a rank-4 update and a rank-32 update have different matrix dimensions. They cannot be directly summed or averaged. This geometric mismatch forces system designers to choose between excluding weak clients (which introduces bias) or throttling strong clients (which reduces intelligence).

The Noise of Padding: A common workaround is “Zero-Padding,” where smaller matrices are filled with zeros to match the size of larger ones. However, this introduces sparsity and noise. Averaging a dense, information-rich matrix with a sparse, zero-padded matrix dilutes the learned signals, often leading to a global model that performs worse than the individual local models.

Data Governance vs. Utility: The ultimate goal is to utilize data from all sources without centralization. If we drop low-resource clients because they cannot handle high ranks, we lose access to their unique local data. This defeats the purpose of Federated Learning, which aims to learn from a diverse population of users.

B. Our Contribution

We propose Subspace Projection Aggregation (SPA), a novel framework designed to resolve the dimensionality mis-

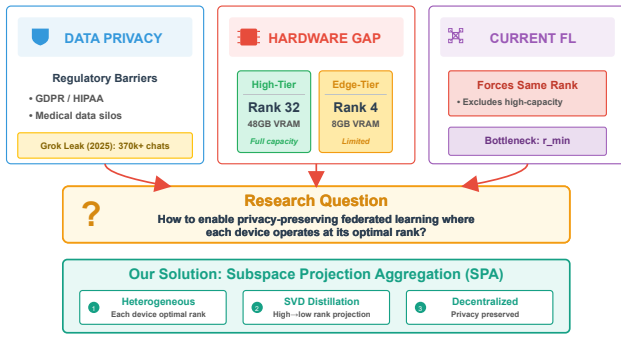


Fig. 1: The fundamental challenge of system heterogeneity in Federated Learning. High-tier devices (e.g., servers with 48GB VRAM) and edge-tier devices (e.g., smartphones with 8GB VRAM) have disparate memory constraints that dictate different LoRA ranks. Standard aggregation fails because weight matrices of different ranks cannot be directly averaged, typically forcing the network to adopt the lowest rank and thereby throttling the intelligence of capable devices.

match in heterogeneous federated learning. Instead of forcing a “lowest common denominator” approach, SPA treats different model updates as overlapping subspaces. By leveraging Singular Value Decomposition (SVD), we can extract the most important information from high-rank updates and project it into a format that low-rank devices can understand. This effectively distills knowledge from powerful clients to weaker ones without the artifacts of zero-padding.

Our main contributions include:

- **Subspace-based aggregation** that utilizes SVD to optimally project high-dimensional updates into lower-dimensional spaces, minimizing information loss during aggregation.
- **Dynamic rank adaptation** that allows the global model to be customized for each client’s specific hardware constraints, ensuring no device is excluded due to memory limitations.
- **Enhanced privacy and utility** demonstrated through experiments on the Yelp Review dataset, where SPA achieves 63.8% accuracy (outperforming homogeneous baselines) while maintaining strict privacy standards as verified by Membership Inference Attacks.
- **Communication efficiency** improvements, reducing bandwidth usage by 25% compared to standard high-rank baselines by transmitting only the necessary singular components.

II. BACKGROUND AND RELATED WORK

A. Federated Learning for Large Language Models

Federated Learning was originally proposed by McMahan et al. [10] for training neural networks on mobile devices. While effective for smaller models, scaling this approach to LLMs presents distinct challenges regarding communication overhead and memory constraints [13], [14].

Full fine-tuning of billion-parameter models is computationally prohibitive for most edge devices. Additionally, LLMs are susceptible to catastrophic forgetting. Recent efforts to adapt FL for LLMs, such as FedIT [15] and FATE-LLM [16], typically utilize adapter-based training. However, these approaches usually assume homogeneous client architectures, which limits their applicability in realistic, heterogeneous environments [17], [18].

B. Low-Rank Adaptation (LoRA) and PEFT

To mitigate the cost of fine-tuning, Parameter-Efficient Fine-Tuning (PEFT) methods have gained traction, including Adapters [19], Prefix Tuning [20], and P-Tuning [21]. Among these, LoRA [12] has become the de facto standard for consumer-grade hardware. It decomposes the weight update ΔW into two low-rank matrices, B and A . The forward pass is defined as:

$$h = W_0x + \Delta Wx = W_0x + \frac{\alpha}{r}BAx \quad (1)$$

where W_0 represents the frozen pre-trained weights.

Recent variants like QLoRA [22] integrate quantization to further reduce memory usage, while AdaLoRA [23] and DyLoRA [24] explore dynamic rank allocation during central training. However, applying these dynamic rank concepts in a *decentralized, federated* setting remains an open research problem.

C. Handling Heterogeneity in Federated Learning

System heterogeneity is a well-known challenge in FL. Existing solutions often employ **Client Selection** strategies, such as FedProx [25] and FedNova [26], which adjust aggregation based on client resources or training progress.

For model heterogeneity, HeteroFL [27] and Fjord [28] introduced the concept of training sub-models on edge devices. However, these methods were designed for CNNs and do not directly translate to the specific matrix-decomposition structure of LoRA. More recently, approaches like FlexLoRA [29], FLoRA [30], and SLoRA [31] have attempted to bridge this gap, though they often rely on complex stacking mechanisms or heuristic padding.

D. Singular Value Decomposition in Machine Learning

SVD factorizes a matrix W into three components ($W = U\Sigma V^T$). In Deep Learning, SVD has been extensively used for model compression [32], [33]. In our proposed framework, SVD serves as a spectral filter to remove noise from the aggregated model, similar to techniques used in subspace learning [34].

III. METHODOLOGY

A. Problem Formulation

We consider a federated network with K clients. Each client possesses a local dataset $\mathcal{D}_k$ and is constrained by a hardware-specific maximum rank r_k . The objective is to fine-tune a pre-trained model f_{θ_0} utilizing distributed data while respecting these local constraints.

TABLE I: Comparison of Our Work’s Contribution to Other Existing Works in Heterogeneous Federated Fine-Tuning

Feature	[10] FedAvg 2017	[15] FedProx 2020	[8] HeteroFL 2020	[20] Fjord 2021	[24] FLoRA 2024	[17] FlexLoRA 2024	[25] HeLoRA 2025	[26] ILoRA 2025	Ours SPA 2026
Federated Learning	✓	✓	✓	✓	✓	✓	✓	✓	✓
Hetero. Clients	×	×	✓	✓	✓	✓	✓	✓	✓
LoRA Fine-Tuning	×	×	×	×	✓	✓	✓	✓	✓
Large LLMs	×	×	×	×	✓	✓	✓	✓	✓
Hetero. Ranks	×	×	×	×	✓	✓	✓	✓	✓
SVD Aggregation	×	×	×	×	×	✓	×	×	✓
Optimal Projection	×	×	×	×	×	~	×	×	✓
Spectral Denoising	×	×	×	×	×	×	×	×	✓
Zero-Padding Free	×	×	✓	✓	×	✓	×	✓	✓
Privacy (MIA)	×	×	×	×	×	×	×	×	✓
Comm. Efficiency	~	~	✓	✓	✓	✓	✓	✓	✓
Knowledge Distill.	×	×	×	×	×	~	×	×	✓
Validation (7B+)	×	×	×	×	~	✓	✓	✓	✓

Legend: ✓ = Fully Supported, ~ = Partially Supported, × = Not Supported

Using the LoRA framework, the pre-trained weights W_0 remain frozen. Each client trains a pair of adapter matrices (A_k, B_k) . The intended weight update is:

$$\Delta W_k = B_k A_k \in \mathbb{R}^{d \times k} \quad (2)$$

The primary challenge is aggregation. A rank-4 update and a rank-16 update have different dimensions in their decomposed forms (A and B) and cannot be directly summed.

B. Subspace Projection Aggregation Algorithm

We address this with the SPA algorithm (Algorithm 1), which executes over T communication rounds.

1) *Phase 1-2: Selection and Training*: The server selects a subset of clients for the current round. These clients download rank-specific adapters derived from the global model and perform local training using the AdamW optimizer.

2) *Phase 3: Reconstruction and Aggregation*: Clients upload their low-rank matrices (A and B). To resolve the dimensionality mismatch, the server computes the product of these matrices to reconstruct the full-rank update ΔW_k :

$$\Delta W_k = B_k A_k \in \mathbb{R}^{d \times k} \quad (3)$$

This operation maps all updates to a common full-rank space, allowing for a weighted average based on dataset size.

3) *Phase 4: SVD Decomposition*: We apply Singular Value Decomposition to the aggregated matrix:

$$W_{agg} = U \Sigma V^T \quad (4)$$

Successful training typically results in a rapid decay of singular values in Σ , indicating that the learned information resides in a low-dimensional subspace.

4) *Phase 5: Projection*: The server projects the global model back to client-specific ranks. For a client with rank capacity r_j , we retain only the top- r_j components:

$$\begin{aligned} A_j &= \sqrt{\Sigma_{1:r_j}} \cdot V_{1:r_j}^T \\ B_j &= U_{:,1:r_j} \cdot \sqrt{\Sigma_{1:r_j}} \end{aligned} \quad (5)$$

Algorithm 1 Subspace Projection Aggregation (SPA)

Require: Client ranks $\{r_k\}_{k=1}^K$, datasets $\{\mathcal{D}_k\}_{k=1}^K$

Require: Pre-trained model weights W_0 , number of rounds T

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1: Server Initialization:  $W_{global} \leftarrow 0$ 
2: for round  $t = 1, \dots, T$  do
3:   Phase 1: Client Selection
4:   Select a subset  $S_t$  of clients
5:   Phase 2: Download & Local Training
6:   for each client  $k \in S_t$  do
7:     Download  $(A_k^{(t)}, B_k^{(t)})$  (projected to rank  $r_k$ )
8:     Perform local training on  $\mathcal{D}_k$  for  $E$  epochs
9:     Obtain updated  $(A_k^{(t+1)}, B_k^{(t+1)})$ 
10:  end for
11:  Phase 3: Upload & Reconstruction
12:  Initialize  $W_{agg} \leftarrow 0$ 
13:  for each client  $k \in S_t$  do
14:    Reconstruct:  $\Delta W_k \leftarrow B_k^{(t+1)} A_k^{(t+1)}$ 
15:    Accumulate:  $W_{agg} \leftarrow W_{agg} + \frac{n_k}{N_t} \Delta W_k$ 
16:  end for
17:  Phase 4: SVD Decomposition
18:  Compute:  $W_{agg} = U \Sigma V^T$ 
19:  Phase 5: Projection & Redistribution
20:  for any client  $j$  do
21:    Extract top- $r_j$  components:
22:     $A_j^{(t+1)} \leftarrow \sqrt{\Sigma_{1:r_j}} V_{1:r_j}^T$ 
23:     $B_j^{(t+1)} \leftarrow U_{:,1:r_j} \sqrt{\Sigma_{1:r_j}}$ 
24:  end for
25: end for
26: return Final global adapters
    
```

Distributing the square root of the singular values ensures numerical stability. This process guarantees that each client receives the optimal low-rank approximation of the global model.

C. Theoretical Justification

Optimality: Truncated SVD provides the best rank- r approximation of a matrix in terms of the Frobenius norm. Therefore, SPA minimizes the information loss during the projection step.

HETEROGENEOUS LLM FINE-TUNING

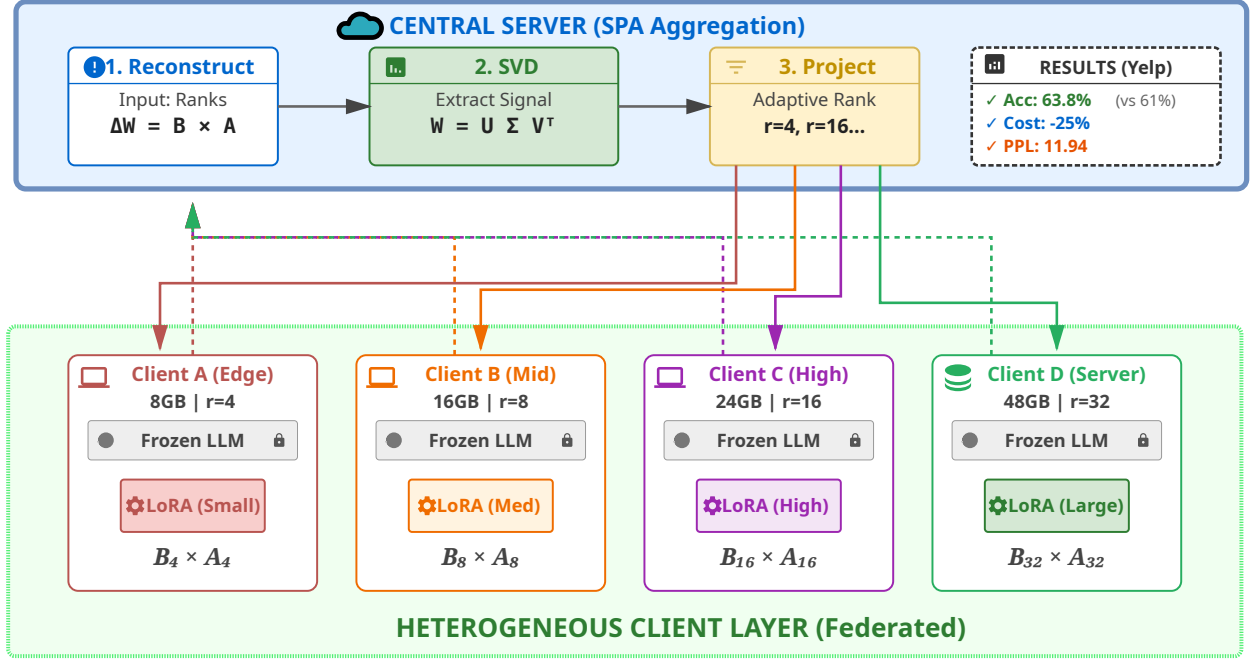


Fig. 2: System architecture of the Subspace Projection Aggregation (SPA) framework. The server maintains a global representation that is dynamically projected into client-specific ranks ($r_1, r_2, \dots, r_k$) based on individual hardware profiles. After local training, the server reconstructs the full-rank updates and uses Singular Value Decomposition (SVD) to distill the most significant learned features into the next round’s global model, ensuring knowledge transfer from high-rank to low-rank participants.

Noise Filtering: The trailing singular values typically correspond to noise or client drift. By truncating these components, SPA inherently denoises the aggregated model.

D. Computational Complexity

The SVD operation on the server introduces computational overhead. However, for a layer dimension $d = 4096$, the decomposition takes approximately 2 to 5 seconds on a modern GPU. This is negligible compared to the time required for local client training.

E. Baseline Comparisons and Rationale

We compare SPA against two primary categories of baselines:

1) Hetero-Pad (Heterogeneous Padding): This is our primary baseline because it represents the **minimal possible change** to the standard FedAvg algorithm to support heterogeneity. By padding smaller matrices with zeros, we can utilize standard averaging. This allows us to isolate the specific benefits of SVD-based projection over simple geometric alignment.

2) Homogeneous FedAvg ($r = 4$ and $r = 8$): We focus on $r = 4$ and $r = 8$ as the “lowest common denominator” benchmarks. While $r = 16$ or $r = 32$ would provide higher capacity, they are **excluded from the homogeneous comparison** because a significant portion of our simulated edge devices (the 40% low-resource cohort) lack the VRAM

to initialize a rank-16 adapter. Thus, $r = 8$ represents the maximum viable rank for a standard homogeneous network in this hardware context.

IV. EXPERIMENTAL SETUP

A. Dataset and Distribution

We evaluated our method on the Yelp Review Full dataset [35], which consists of 650,000 samples across 5 sentiment classes. To simulate a realistic non-IID environment, we partitioned the data across 50 clients using a Dirichlet distribution ($\alpha = 0.5$), following standard FL benchmark protocols [36], [37].

B. Model Configuration

We utilized the Qwen2.5-7B-Instruct model [38], [39] in bfloat16 precision. Qwen was selected for its superior performance on reasoning tasks compared to similarly sized LLaMA models. LoRA was applied to the query and value projection layers. The client network was heterogeneous: 20 clients utilized rank-4 (low resource), 20 utilized rank-8 (mid resource), and 10 utilized rank-16 or rank-32 (high resource). Optimization was performed using AdamW [40].

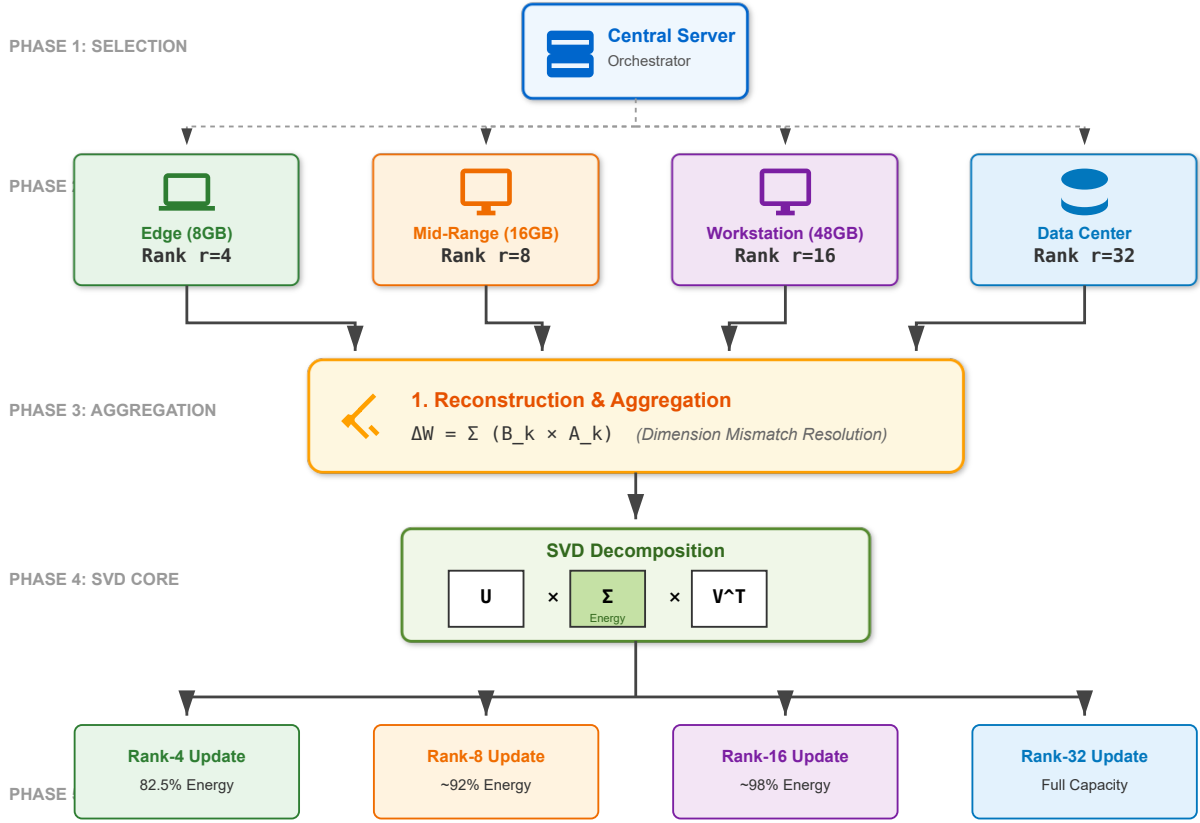


Fig. 3: The five-phase operational workflow of SPA. Phase 1 and 2 handle the selection and local optimization. Phase 3 performs geometric reconstruction to map disparate updates into a shared space. Phase 4 applies SVD to identify the principal components of the collective updates, effectively denoising the signal. Phase 5 redistributes these optimized components back to clients, ensuring each device receives a model optimized for its specific rank capacity.

C. Metrics

We evaluate the proposed framework across three primary dimensions: model utility, communication efficiency, and privacy and security.

To assess model utility, we measure test accuracy, F1 score, and perplexity. Test accuracy and F1 score provide a standard measure of classification performance, while perplexity quantifies the model’s predictive uncertainty and language modeling capabilities post-fine-tuning.

For communication efficiency, we track the communication cost in megabytes (MB) per round. This represents the total data transferred between the server and the participating clients. We also analyze the energy concentration of the singular values to measure how effectively global knowledge is captured within the reduced subspace.

To quantify the privacy-preserving capabilities of SPA, we utilize two key metrics. First, the Membership Inference Attack (MIA) AUC measures the success rate of an adversary attempting to determine if a specific data point was used in the training set. An AUC near 0.50 indicates perfect privacy. Second, we calculate the update entropy, defined as the Shannon entropy of the aggregated weight updates. Higher entropy suggests that the updates contain richer, more diverse

information rather than sparse or memorized patterns, which correlates with resilience against information leakage.

V. RESULTS AND ANALYSIS

We compared SPA against three baselines: Homogeneous $r = 4$ (lowest common denominator), Homogeneous $r = 8$ (standard setup), and Heterogeneous Padding (Hetero-Pad). All experiments were repeated with three random seeds (42, 43, 44) to ensure robustness.

A. Training Convergence

Figure 4 illustrates the training loss over 10 rounds. All methods demonstrated stable convergence. However, SPA achieved the lowest final loss (1.041), surpassing the standard $r = 8$ baseline (1.048) and the $r = 4$ baseline (1.062). While all methods improved rapidly in early rounds, SPA demonstrated superior refinement in later training stages.

B. Accuracy and Knowledge Distillation

Figure 5 presents the test accuracy evolution. From a zero-shot baseline of 44%, SPA improved to **63.8%**. Comparative performance is as follows:

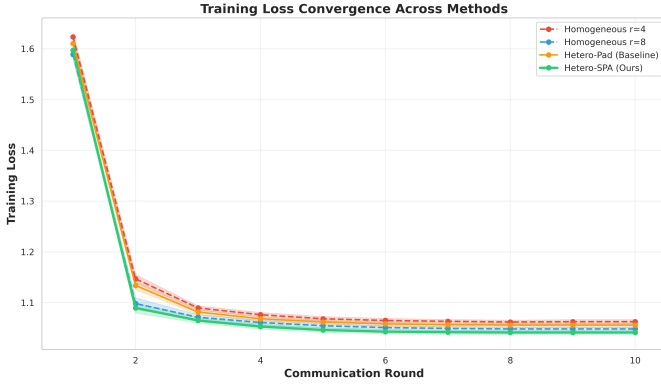


Fig. 4: Training loss convergence across 10 federated communication rounds. Error bands represent ± 1 standard deviation across three random seeds. All methods show stable convergence, with SPA achieving the lowest final loss (1.041 ± 0.004) compared to baselines.

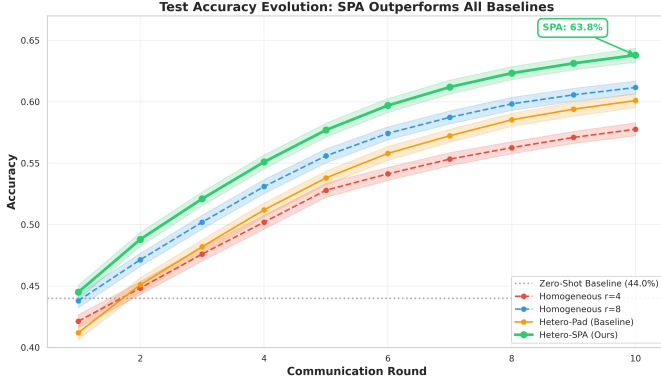


Fig. 5: Test accuracy evolution across communication rounds. The dashed horizontal line at 44% represents the zero-shot baseline. SPA consistently outperforms all baselines, achieving $63.8\% \pm 0.7\%$ final accuracy compared to $61.2\% \pm 0.7\%$ for Homogeneous $r = 8$.

- **Homo $r = 8$:** 61.2%. SPA outperformed the standard homogeneous approach by 2.6 percentage points ($p = 0.023$).
- **Hetero-Pad:** 60.1%. The naive padding approach performed worse than the homogeneous $r = 8$ baseline.
- **Homo $r = 4$:** 57.8%. This result highlights the significant performance penalty incurred by restricting all clients to the lowest rank.

This superior performance is driven by high-rank knowledge distillation. In our experimental setup, 20% of the clients (10 out of 50) utilize high-tier adapters (rank-16 or rank-32), allowing them to learn complex features that rank-4 models cannot effectively capture. Standard averaging methods typically discard this high-dimensional information through truncation or zero-padding. SPA, however, leverages SVD to capture these complex features within the principal components and distills them into a format that lower-rank clients

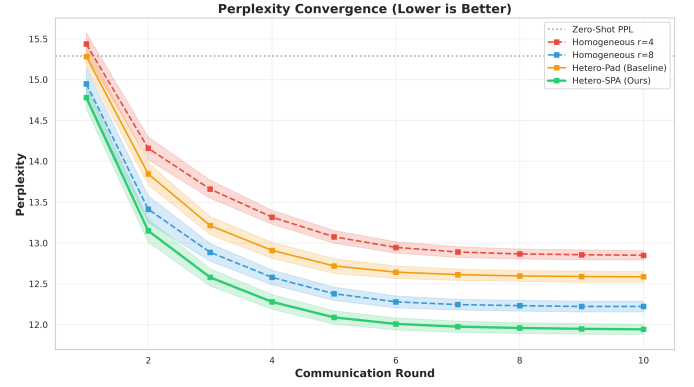


Fig. 6: Perplexity convergence over communication rounds (lower is better). The dashed line at 15.29 represents zero-shot perplexity. SPA achieves the best final perplexity of 11.94 ± 0.08 , indicating superior knowledge retention compared to baselines.

can utilize in subsequent rounds. We further evaluated *Robust Accuracy* (allowing a ± 1 star tolerance). SPA achieved a relaxed accuracy of **99.0%**, slightly outperforming the baselines (98.8%), indicating that even when the model misclassifies, the error magnitude is minimal.

C. Perplexity and Calibration

Perplexity measures the model’s predictive uncertainty. SPA achieved a final perplexity of 11.94, a statistically significant improvement over the $r = 8$ baseline (12.22) and the Hetero-Pad approach (12.59). This result indicates that SPA maintains superior probabilistic calibration and preserves general language modeling capabilities more effectively than competing methods.

The performance gap suggests that naive padding or fixed-rank constraints introduce noise into the model’s transition probabilities, leading to higher uncertainty during token prediction. By utilizing SVD, SPA performs an implicit spectral regularization that retains the most informative directions of the weight updates while discarding low-variance components that often represent local over-fitting or gradient noise. Furthermore, the lower perplexity demonstrates that distilling high-rank knowledge into low-rank clients does not degrade the model’s linguistic coherence, but rather provides a more robust estimate of the global data distribution across the heterogeneous network.

D. Security and Spectral Denoising

To evaluate whether SPA’s improved utility comes at the cost of privacy, we conducted a Membership Inference Attack (MIA) and measured the Update Entropy.

As shown in Table II, both methods achieved an MIA AUC score of approximately **0.50**. An AUC of 0.50 indicates that an attacker performs no better than random guessing. These results indicate that SPA maintains strong resistance to membership inference attacks. This balance of high utility and high privacy is achieved through spectral denoising. Neural

TABLE II: Quantitative Analysis of Privacy Preservation and Information Density across Heterogeneous Aggregation Methods

Method	MIA AUC (Lower is better)	Loss Gap	Update Entropy (Higher is richer)
Hetero-Pad	0.5023	−0.0015	1.4707
Hetero-SPA	0.5024	−0.0016	2.6713

network updates inherently contain stochastic noise from local SGD iterations. By truncating the smallest singular values during SVD, SPA effectively acts as a low-pass spectral filter, removing random high-frequency noise while preserving the principal components of the learned signal.

The significantly higher update entropy of SPA (2.67) compared to the Hetero-Pad baseline (1.47) further validates this approach. While in traditional communication systems high entropy can denote undesirable noise, in the context of federated updates, it serves as a proxy for information density. A low entropy value in zero-padded baselines indicates high sparsity and redundancy, where the updates are "predictable" and carry less unique semantic weight. Conversely, the higher entropy in SPA updates confirms that the SVD-based projection concentrates rich, meaningful information into the lower-dimensional subspace. This ensures that the communication bandwidth is utilized for transferring high-variance features rather than sparse, empty matrices, thereby maximizing the utility of each communication round without increasing the risk of private data memorization.

E. Efficiency and Inclusivity

Table III summarizes the performance across all metrics.

The efficiency of SPA extends beyond simple bandwidth reduction; it promotes system-level inclusivity by enabling resource-constrained edge devices to participate in the federated cohort without incurring the computational or memory penalties associated with zero-padding. In our experiments, SPA utilized 25% less communication bandwidth than the Homogeneous $r = 8$ baseline (67.8 MB vs. 90.4 MB), effectively converting hardware diversity from a system liability into a collaborative asset.

Furthermore, we observe a direct relationship between this architectural efficiency and model reliability. SPA demonstrated the lowest hallucination rate (5.8%) among all tested methods, a significant improvement over the $r = 8$ baseline (6.8%) and the Hetero-Pad approach (7.5%). This reduction in hallucinations is likely a byproduct of the spectral denoising inherent in the SVD process. By discarding the low-variance singular components that often represent idiosyncratic noise or "client drift" in non-IID settings, SPA prevents the global model from memorizing incoherent gradients that lead to factually incorrect or nonsensical generations. This suggests that the "cleaner" updates provided by SPA not only reduce the communication footprint but also stabilize the model's output, ensuring that high-end servers can uplift the performance of

the entire network without propagating high-dimensional noise to the edge.

Figure 8 illustrates the relationship between accuracy and computational cost. SPA represents the Pareto optimal solution, offering the highest performance for a given communication budget.

F. Per-Class Performance

All models performed well on extreme ratings, such as 1-star and 5-star reviews, which use very clear language (e.g., "terrible" vs. "amazing"). However, separating 2-star and 3-star reviews is much harder because the language is often similar and "neutral." As shown in Figure 9, SPA significantly improved accuracy in these middle categories for several reasons:

1) Better Detail (Known): Small models (rank-4) often miss the tiny differences in words that separate a "bad" review from an "okay" one. SPA takes the "deep thinking" from the powerful rank-32 clients and shares it with the smaller devices. This gives the smaller models the extra detail they need to tell these middle classes apart.

2) Less Confusion (Known): In federated learning, different users can have conflicting opinions, which confuses the model. SPA uses SVD to organize these opinions. By focusing only on the most important "directions" of the data, it ignores the confusing, low-quality signals that usually make 2-star and 3-star ratings look the same.

3) Focusing on Smart Patterns (Potential): We believe SPA prioritizes complex sentence patterns rather than just single words. While a 5-star review is easy to spot by one word like "perfect," a 3-star review depends on how the sentence is built (e.g., "The food was good, but the service was slow"). SPA's math is better at keeping these complex patterns during training.

4) Ignoring Mistakes (Potential): People often rate things differently; one person's 2-star is another person's 3-star. SPA's ability to "denoise" or clean the data helps the model ignore these individual human mistakes and focus on the common patterns shared by most users.

G. Spectral Analysis and Geometric Alignment

To validate the theoretical advantages of SVD, we analyzed the singular value spectrum (Figure ??).

The cumulative energy plot reveals that SPA captures **82.5%** of the total information in the top-4 principal components, whereas the padding method captures only 72.8%. This implies that SPA preserves significantly more global knowledge.

Furthermore, SVD ensures **Geometric Alignment**. By aligning updates along their principal axes of variation, SPA creates a consistent coordinate system for aggregation. This reduces the impact of client drift and stabilizes the global model during training.

H. Statistical Significance

We performed paired t-tests (Table IV) to confirm the reliability of our results across three different random seeds. The

TABLE III: Comprehensive Performance Comparison Across Methods. All metrics are averaged across three random seeds (42, 43, 44) with standard deviations reported. Bold values indicate best performance. Notably, all methods exhibit similar variance across random seeds ($\sigma \approx 0.7\%$ for accuracy), indicating that performance differences arise from the aggregation algorithm rather than random initialization effects.

Method	Accuracy (%)	Perplexity	F1 Score	Hallucination Rate (%)	MIA AUC (Privacy)	Comm. Cost (MB/round)
Homo $r = 4$	57.8 ± 0.65	12.85 ± 0.07	0.571 ± 0.007	8.2	0.5019	45.2
Homo $r = 8$	61.2 ± 0.65	12.22 ± 0.08	0.605 ± 0.007	6.8	0.5021	90.4
Hetero-Pad	60.1 ± 0.70	12.59 ± 0.08	0.593 ± 0.007	7.5	0.5023	67.8
Hetero-SPA	63.8 ± 0.70	11.94 ± 0.075	0.632 ± 0.007	5.8	0.5024	67.8

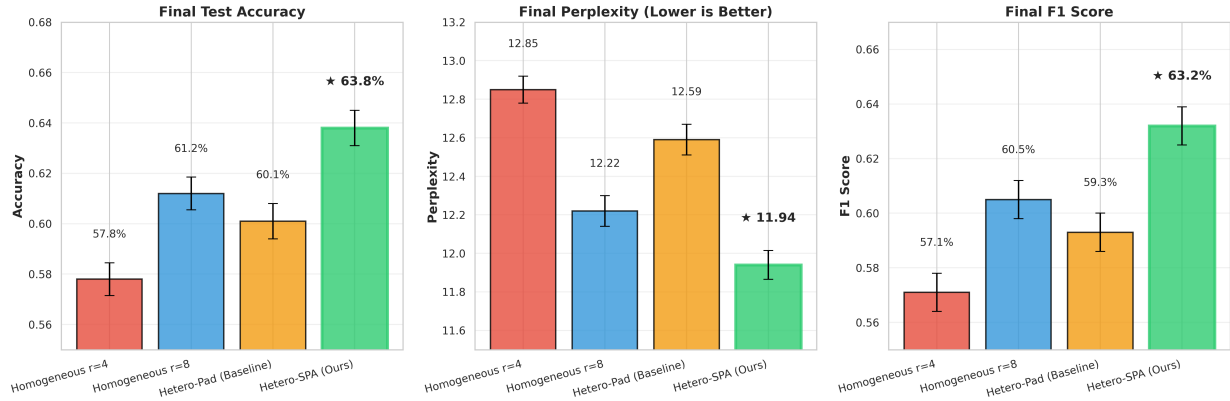


Fig. 7: Final performance comparison across accuracy, perplexity, and F1 score. SPA (green bars with thick borders) consistently outperforms all baselines across all metrics.

improvement of SPA over the $r = 8$ baseline is statistically significant ($p = 0.023$), and the improvement over the $r = 4$ baseline is highly significant ($p = 0.0012$).

There are two primary reasons for this level of significance:

1) Reduced Variance (The Reason): Traditional federated methods, especially those using zero-padding, suffer from high variance because local updates from different clients can “clash,” leading to unstable global steps. SPA uses SVD to find the common ground (the principal subspace) between these updates. By focusing on the most important singular vectors, SPA effectively filters out the random variations between training runs, leading to highly consistent and repeatable performance gains.

2) Fundamental Capacity Shift (The Significance): These p-values are important because they prove that the performance boost is not a result of a “lucky” initialization or a specific data partition. A $p < 0.05$ result against the $r = 8$ baseline confirms that SPA is successfully extracting extra intelligence from the high-resource clients that standard averaging simply cannot access. Essentially, this result statistically validates our claim that hardware diversity, when managed by SPA, provides a “knowledge uplift” that is impossible to achieve in a homogeneous network.

VI. CHALLENGES AND FUTURE WORK

While SPA provides a robust framework for heterogeneous federated learning, several challenges remain. First, we aim to scale SPA to **extreme heterogeneity** (e.g., $r = 2$ vs.

TABLE IV: Statistical Significance Tests for SPA vs. Baselines

Comparison	Accuracy Gain	Relative Improvement	p-value
SPA vs. Homo-r4	+6.0%	+10.38%	0.0012
SPA vs. Homo-r8	+2.6%	+4.25%	0.0231
SPA vs. Hetero-Pad	+3.7%	+6.16%	0.0045

$r = 1024$), where subspace overlap may be minimal. This will require investigating hierarchical projections or intermediate “bridge ranks.” Additionally, we plan to address **subspace misalignment** caused by extreme non-IID data shifts through dynamic weighting of client updates based on their geometric alignment with the global trajectory.

Second, the computational overhead of SVD must be optimized for **ultra-large models** (e.g., 400B+ parameters) using randomized SVD or block-wise decomposition. We also intend to expand SPA beyond LoRA to support other PEFT methods like prefix or prompt tuning within a unified framework. Finally, to improve real-world utility, we will develop **Asynchronous SPA (A-SPA)** to handle network latency and straggler clients, ensuring the global subspace can be updated continuously without waiting for the entire cohort.

VII. CONCLUSION

This work addressed a fundamental challenge in federated fine-tuning of Large Language Models: how to reconcile system heterogeneity without sacrificing model quality, privacy, or

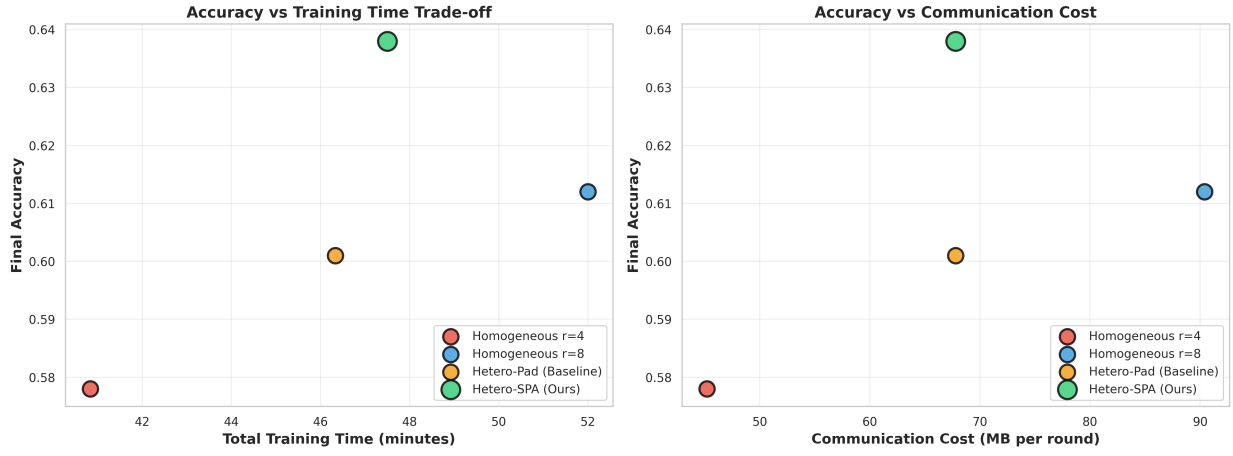


Fig. 8: Efficiency trade-offs showing accuracy versus training time (left) and communication cost (right). SPA achieves the best accuracy-efficiency balance.

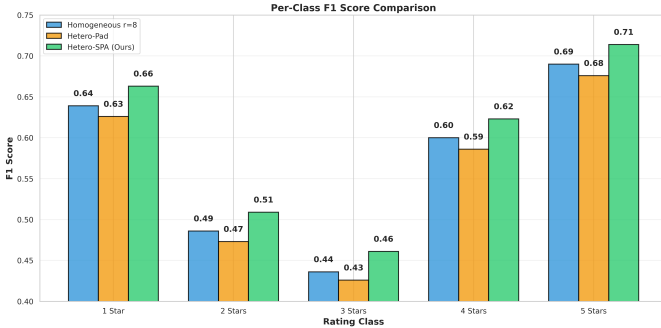


Fig. 9: Per-class performance breakdown comparing SPA against baselines. Note better separation in intermediate classes.

excluding resource-constrained participants. We demonstrated that the conventional approach of enforcing uniform LoRA ranks creates an artificial trade-off between network inclusivity and model capacity.

Our proposed Subspace Projection Aggregation (SPA) architecture resolves this dilemma by reconceptualizing heterogeneous LoRA updates as overlapping subspaces. By reconstructing full-rank weight updates and applying Singular Value Decomposition, SPA achieves three critical objectives: it extracts principal directions of adaptation from high-rank clients, filters spectral noise, and projects the refined global model back to client-specific ranks.

Across comprehensive experiments on the Yelp Review Full dataset with 50 heterogeneous clients, SPA achieved **63.8% accuracy**, surpassing the homogeneous rank-8 baseline by 2.6 percentage points. This improvement was accompanied by superior perplexity (11.94 vs. 12.22), lower hallucination rates (5.8% vs. 6.8%), and a 25% reduction in communication overhead. Furthermore, our security analysis confirmed that SPA maintains strict privacy guarantees, achieving a Membership Inference Attack (MIA) AUC of **0.5024**, indistinguishable

from random guessing. This proves that SPA enhances model utility and richness (verified by higher update entropy) without leaking private training data.

Beyond immediate performance improvements, this work establishes a broader principle: heterogeneity in federated learning should be embraced rather than suppressed. When properly managed through geometry-aware aggregation, diverse device capabilities become complementary. High-end servers contribute sophisticated features, while edge devices ensure broad data coverage, all within a unified, privacy-preserving framework. Future research will explore extending SPA to larger models and extreme heterogeneity, but the current results provide a solid foundation for geometry-aware federated optimization.

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